

ECON

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- Lecture 1–4
- Lecture 5 Economic Optimization
- Lecture 6 Demand Analysis
- Lecture 7 Demand Estimation
- Lecture 8 Demand Forecasting
- Lecture 9 Cost Analysis
- Lecture 10 Production Analysis
- Lecture 11–12 Market Structure
- Lecture 13 Pricing Practice
- Lecture 14 Public Policy
- Lecture 15 Risk and Uncertainty
- Lecture 16 Capital Budgeting

- What to produce
- How to produce
- For whom to produce
- Who makes decision

- Capitalist
- Communist
- Socialist
- Adam Smith: The Wealth of Nations, Self Interest, Free Market, Invisible Hand Small Government
- Karl Marx

Macro

Micro

- Macro Economics
- Micro Economics
- 4Ps Product Price Place Promotion
- 5Cs Character Capability Capital Condition Collateral

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2-5

- Stable Growth
- Stable Price
- Full Employment
- Fair Distribution of Income
- Efficiency
-

GNP

GDP

GDP = GNP -

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$$Y = C + I + G + X - M$$

Investment

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GNP

Nominal Real GDP

$$\text{Real GDP} = (\text{Nominal GDP} / \text{Price Index}) \times 100$$

- CPI
- PPI

: GDP

		Price Index	Nominal GDP	Real GDP
1	10	50	20,000	40,000
2	20	100	25,000	25,000
3	30	150	30,000	20,000

Consumption Saving

Multiplier

$$C = a + bY$$

$$MPC = \Delta C / \Delta Y$$

$$MPS = \Delta S / \Delta Y$$

$$MPC + MPS = 1$$

$$Y = [1/(1-b)](a + I + G + X - M)$$

$$\Delta Y = [1/(1-b)](\Delta a + \Delta I + \Delta G + \Delta X - \Delta M)$$

Lump Sum Tax: $C = a + b(Y-T)$ T $-b/(1-b)$ Balanced Budget
Multiplier $\Delta G = \Delta T$ 1

$$: Y = [1/(1-b+bt)](a + I + G + X - M)$$

Aggregate Demand

Y M I

Multiplier

2-3

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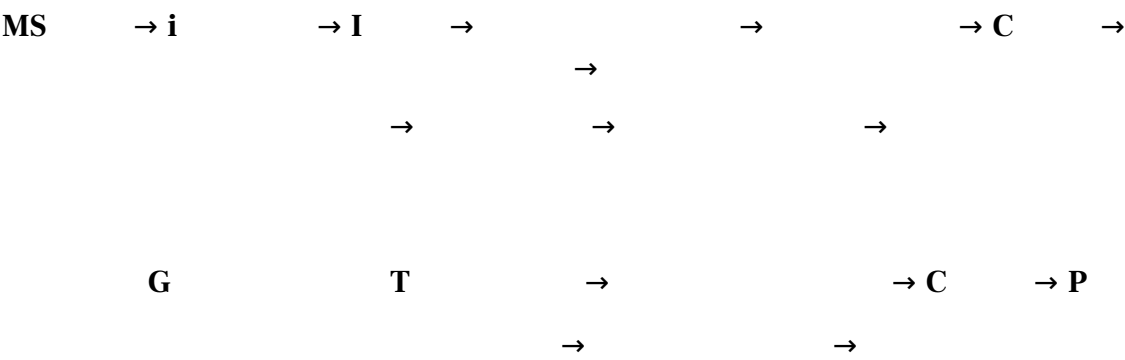
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2-3

- Creeping Inflation 1% 3%
- Double Digit Inflation 10% 99%
- Hyper Inflation 100%

Core Inflation Rate

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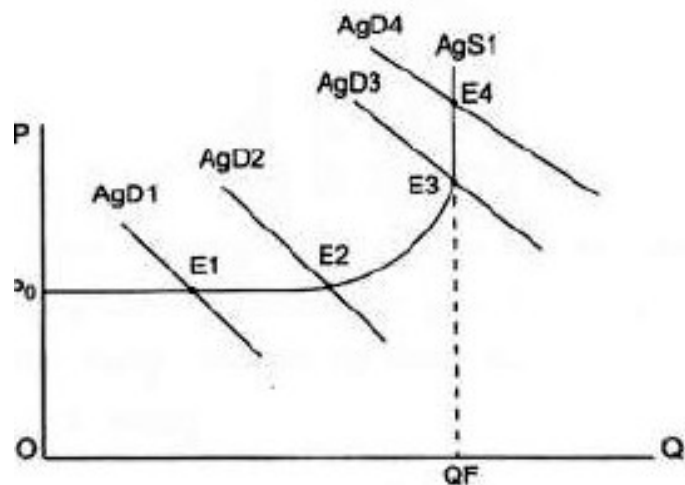
Demand Pull Inflation

Aggregate Demand

Full Employment

Aggregate Demand

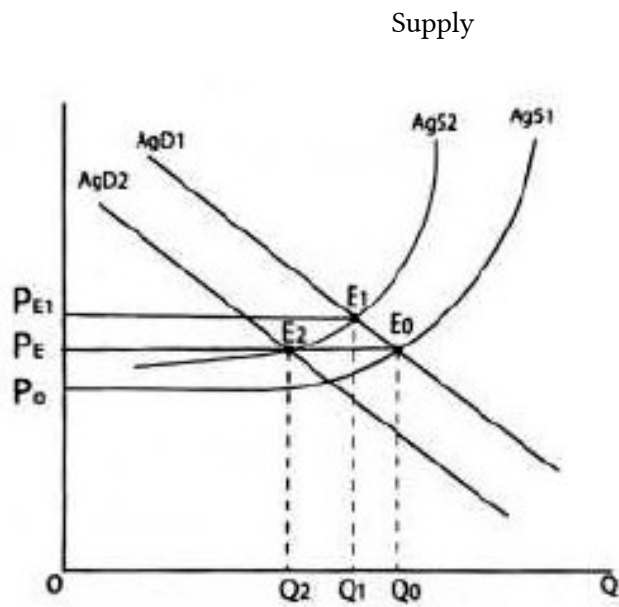
Full Employment



Demand Pull Inflation

Employment	E3	E4	E2	E3	Full
	Aggregate Demand		Money Supply		

Cost Push Inflation



Cost Push Inflation

AgS1	AgS2	E0	Aggregate Supply
		E1	
Aggregate Demand			Output

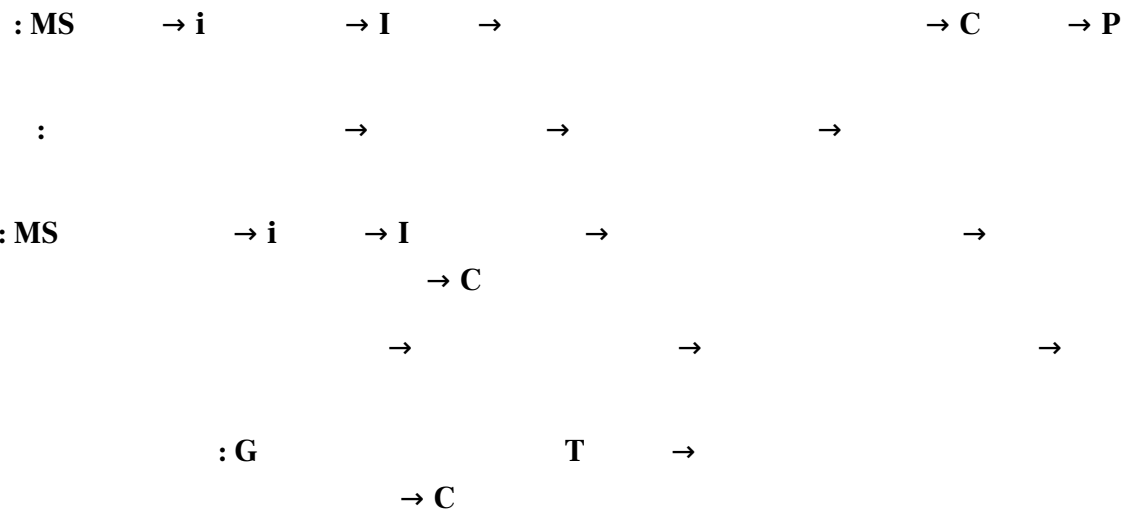
Profit Push Inflation

Fiscal Policy

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Monetary Policy

Money Supply



Financial Market

Financial Market Financial Intermediaries

- Organized Market
- Unorganized Money Market
- Money Market
- Capital Market

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- Quantitative Tools: Required Reserve, Discount Rate, Open Market Operation
 - Qualitative Tools: Selective Interest Control, Selective Credit Control, Moral Suasion
 - Money Supply: Required Reserve, Discount Rate
 - Money Supply: Required Reserve, Discount Rate
- Required Reserve Money Multiplier Excess Reserve
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$\frac{100}{900} \times 100 = 11.1\%$

- M1 Narrow Money:
- M2 Broad Money: M1
- M2a: M2
- M3:

Discount Rate Federal Funds Rate Policy Interest Rate

- Federal Discount Rate Federal Reserve
- Federal Funds Rate Discount Window
- Discount Rate Federal Funds Rate
- Discount Rate Fed
- Policy Interest Rate
- Policy Rate

Absolute Advantage
 Comparative Advantage

Exchange Rate

- Foreign Exchange Market Forex, FX, Currency Market
- International Reserves
- Special Drawing Right SDR
- Fixed Exchange Rate Devaluation
- Flexible/Floating Exchange Rate Demand Supply
- Depreciation Appreciation
- Dirty Floating Clean Floating
- Semi-floating Basket of Currencies

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$$= eP^* / P$$

Balance of Payments

- Current Account Trade Account, Service Account Transfer Account
- Capital Account

- Country Risk, Currency Risk Credit Risk

$$BP = NX(Y, eP^*/P) + CF(i - i^*)$$

NX

$\frac{Y}{eP^*/P}$

CF

Business Cycle

Trough → Expansion → Boom → Recession →

Trough

4 7 15 5 10 19

4

- Medium of Exchange
- Unit of Account
- Store of Value

Commodity Money

Intrinsic Value

Fiat Money

Block Chain

Bitcoin

- Bitcoin . . 2009 Peer to Peer Electronic Cash System
- Bitcoin 21
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Smart Contract

1,500 Bitcoin, Ethereum, Ripple, Bitcoin Cash

Litecoin Ethereum Smart Contract

Block Chain Ether

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$$\text{Value of the Firm} = \sum [(R_t - C_t) / (1+r)^t]$$

- Total Cost = Total Fixed Cost + Total Variable Cost
- Average Fixed Cost = TFC/Q , Average Variable Cost = TVC/Q , Average Total Cost = TC/Q
- Marginal Cost = $\Delta TC / \Delta Q$
- Law of Diminishing Returns

MC

AVC

ATC

TR

TC

MR = MC

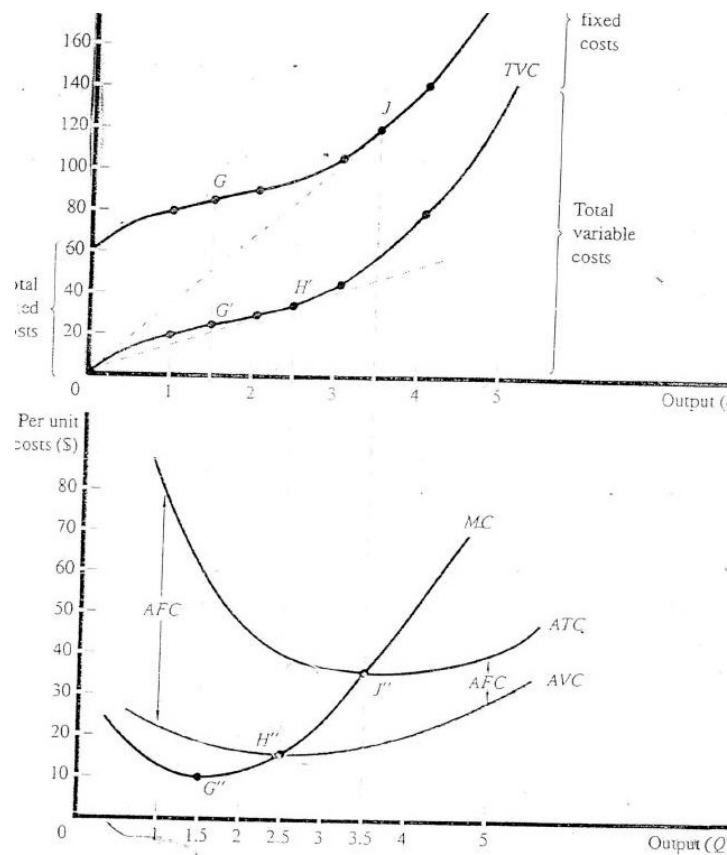


Figure 7-1
 Short-Run Total and Per-Unit Cost Curves

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0

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First-order condition: $df/dx = 0$

Second-order condition: $d^2f/dx^2 > 0$; $d^2f/dx^2 < 0$

Substitution Partial Derivative
Lagrangian Multiplier

Optimization

1 $TR = 100Q - 10Q^2$

$dTR/dQ = 100 - 20Q = 0 \rightarrow Q = 5$

$d^2TR/dQ^2 = -20$

$TR = 100(5) - 10(25) = 250$

2 Partial Derivative $\pi = 80x - 2x^2 - xy - 3y^2 + 100y$

$\partial\pi/\partial x = 80 - 4x - y = 0$

$\partial\pi/\partial y = -x - 6y + 100 = 0$

: $x = 16.52, y = 13.92$ $\pi = 1,356.52$

3 Constrained Optimization by Substitution

x +

y = 12

$x = 12 - y \rightarrow \pi = -4y^2 + 56y + 672$

$d\pi/dy = -8y + 56 = 0 \rightarrow y = 7, x = 5, \pi = 868$ $d^2\pi/dy^2 = -8$

4 $TC = 3x^2 + 6y^2 - xy$ $x + y = 20$

: $x = 20 - y \rightarrow TC = 1,200 - 140y + 10y^2$

$dTC/dy = -140 + 20y = 0 \rightarrow y = 7, x = 13, TC = 710$ $d^2TC/dy^2 = 20$

$LTC = 3x^2 + 6y^2 - xy + \lambda(20 - x - y) \rightarrow x = 13, y = 7, \lambda = 71, TC = 710$

Demand Analysis

Demand /

- Direct Demand
- Derived Demand
- Change in Quantity Demanded
- Change in Demand

Indifference Curve

Budget Line

Indifference Curve

2

Budget Line

Supply

- Equilibrium
- Surplus Excess Supply
- Shortage Excess Demand

- Constant Function Rule
- Power Function Rule
- Sum and Difference Rule
- Product Rule
- Quotient Rule
- Chain Rule

6 1 27

6

Price Floor

Price Ceiling

6

- Equilibrium
- Surplus Excess Supply
- Shortage Excess Demand

Business Profit = TR - Total Accounting Cost

Business Profit = TR - Total Explicit Cost

Explicit Cost

Economic Profit = TR - Total Economic Cost

Economic Profit = TR - (Total Accounting Cost + Opportunity Cost)

$$\text{Economic Profit} = \text{TR} - (\text{Total Explicit Cost} + \text{Total Implicit Cost})$$

6

3

23

7

Elasticity = Percentage Change in Dependent Variable / Percentage Change in Independent Variable

Price Elasticity of Demand E_p = Percentage Change in Q_x / Percentage Change in P_x

Point $E_p = (dQ/dP)(P/Q)$

Arc $E_p = [(Q_2 - Q_1)/(P_2 - P_1)] \times [(P_2 + P_1)/(Q_2 + Q_1)]$

Price Elasticity

$ E_p < 1$	Inelastic
$ E_p = 1$	Unitary Elastic
$ E_p > 1$	Elastic

- E_p
- E_p
- E_p
- E_p
- E_p
- E_p
- $|E_p| < 1$: TR
- $|E_p| = 1$: TR
- $|E_p| > 1$: TR

Income Elasticity of Demand

E_y = Percentage Change in Q_x / Percentage Change in Income

- $E_y > 1$ Superior Goods Cyclical Normal Goods
- $0 \leq E_y \leq 1$ Normal Goods Non Cyclical Normal Goods
- $E_y < 0$ Inferior Goods

Cross Price Elasticity of Demand

E_{xy} = Percentage Change in Q_x / Percentage Change in P_y

- $E_{xy} > 0$ X Y Substitution Goods
- $E_{xy} < 0$ X Y Complementary Goods
- $E_{xy} = 0$ X Y Independent Goods

Demand, TR, MR

$$P = a - bQ \quad TR = P \times Q = aQ - bQ^2$$

$$MR = dTR/dQ = a - 2bQ$$

$$MR = P(1 + 1/E_p)$$

$$MC = MR \quad P = MC/(1 + 1/E_p)$$

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Point Elasticity

$$A = -1$$

$$B = -0.33$$

Arc Elasticity

$$10 \quad 5$$

$$200 \quad 300 = -0.6$$

$$E_p = -2.5 \quad MC = 10 \rightarrow P = 10/(1 + 1/-2.5) = 16.67$$

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8

1. Consumer Interviews

2. Market Experiments

Laboratory Experiment

Actual Market Experiment

3. Regression Analysis

Consumer Interviews

Target Group

Pilot Test

Train Interviewer

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Laboratory Experiment

Actual Market Experiment

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Regression Analysis

Identification Problem

Regression Analysis

1. Specify Variable to be Included in the Model
(Ad, ...)

$$Q_x = f(P_x, P_s, P_c, Y,$$

2. Collecting Data on the Variable

Time Series Data

Cross Section Data

3. Specifying the Form of the Equation

Log

4. Interpret Statistical Results

t F R²

Adjusted R²

5. Interpret the Results of the Model

- t statistic
 - R^2
 - Adjusted R^2
 - F statistic
- Critical Value
 Regression Equation
 Dependent Variable
 Independent Variable
 F
 Critical Value
 Reject H_0

Regression

BLUE Best Minimum Variance Among All Unbiased Estimator, Linear, Unbiased Estimator

Regression Line Error Term Expected Value 0

- Multicollinearity: Independent Variable Correlation
- Heteroskedasticity: Variance Error Term Cross Section Data
- Autocorrelation Serial Correlation: Error Terms

Log Weighted Least Squares
 Dependent Independent Variables
 Model
 Independent
 Variable
 Dependent Variable

$$\text{Adjusted } R^2 = R^2 - [(k - 1)/(n - k)](1 - R^2)$$

k Intercept n

$$F = [\text{Explained Variation}/(k - 1)] / [\text{Unexplained Variation}/(n - k)]$$

$$F = [R^2/(k - 1)] / [(1 - R^2)/(n - k)]$$

$$H_0: b_1 = b_2 = \dots = b_n = 0 \quad H_a: b \neq 0$$

F Critical Value Reject H_0

Dependent Variable

Independent Variables

8 1 4 9 14 22

1–8

Crypto Price floor/Price ceiling

Regression

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Bitcoin

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1		2 3
2	Crypto Bitcoin	4
2	Price floor Price ceiling	equilibrium surplus shortage 6
2		6
3		7
4	Regression	8

- 1.
- 2.
- 3.
- 4.

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- Demand Pull E2 E3 E4
- Cost Push E0 E1 E2
- Profit Push
- MS G T

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- Crypto Block Chain
- Bitcoin Peer to Peer Electronic Cash System 21
- Equilibrium Surplus Shortage
- Economic Profit Explicit Cost Implicit Cost

3

- Own Price Income Cross Price
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4 Regression

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- R^2 Adjusted R^2 F Critical F

3

	Price Elasticity	X	-3	Income Elasticity	1.5	Cross Price Elasticity	X
Y	2	Z	-2				
	-2	Cross Price Elasticity		Z		X	
	Z	Z	1%	X	2%		
	Income Elasticity		1.5	1	Superior Product	Cyclical	
Normal Product							
	Y	1%	Qx	2%	Z	1%	Qx
	0%						
	Cross Price Elasticity		X	Y	2		
			Qx		3 - 1.5 - 2 + 2 = 1.5%		

$$\mathbf{Px} \quad 1\% = (-3)(-1\%) = +3\%$$

$$1\% = (1.5)(-1\%) = -1.5\%$$

$$\mathbf{Py} \quad 1\% = (2)(-1\%) = -2\%$$

$$\mathbf{Pz} \quad 1\% = (-2)(-1\%) = +2\%$$

$$\mathbf{Qx} = +1.5\%$$

7

4

$$Q_x = 50 - 200P_x + 50P_s + 5A - 2P_c$$

t statistic Critical Value t 2.0

	t		
50	2.5		
$P_x - 200$	3.0	X	1 Q_x 200
$P_s + 50$	2.7	S	1 Q_x 50
A + 5	1.5	5	1 Q_x t
$P_c - 2$	3.0	C	1 Q_x 2

R² Adjusted R²

R² = 0.8 X
80% 20%

Adjusted R² = 0.6 60%

F Statistic

F = 5 Critical F = 3 Reject H0

Variable Independent Variables Dependent